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(54) **DISPLAY PANEL WITH CVD
ENCAPSULATION LAYERS, DISPLAY
APPARATUS COMPRISING THE
ENCAPSULATION LAYERS, AND METHOD
FOR MANUFACTURING THE DISPLAY
PANEL**

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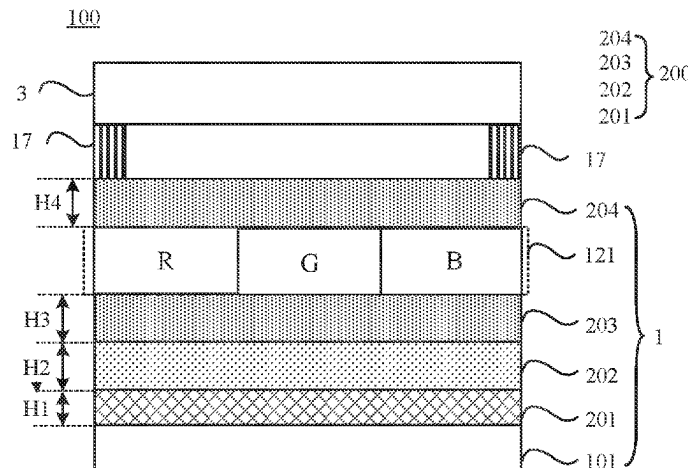
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ABSTRACT

A display panel includes a display substrate and an encapsulation layer. The display substrate includes: a base substrate; and a plurality of light-emitting devices disposed on the base substrate, the plurality of light-emitting devices being configured to emit light. The encapsulation layer is located on a side of the light-emitting devices away from the base substrate, and is configured to encapsulate the light-emitting devices. The encapsulation layer includes: a second encapsulation layer located on the side of the light-emitting devices away from the base substrate, and made of a silicon nitride material; and a third encapsulation layer located on a side of the second encapsulation layer away from the

(Continued)



light-emitting devices, and made of a silicon carbonitride material or a silicon carbide material.

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 See application file for complete search history.

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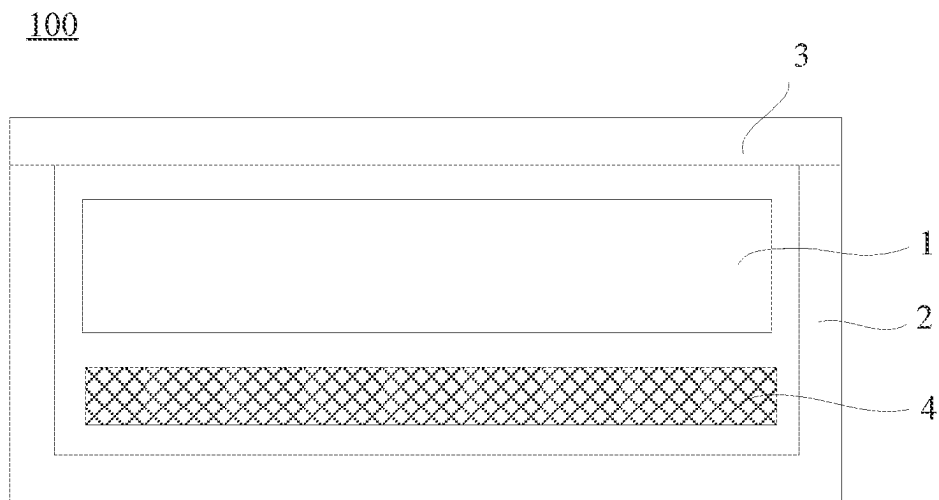


FIG. 1

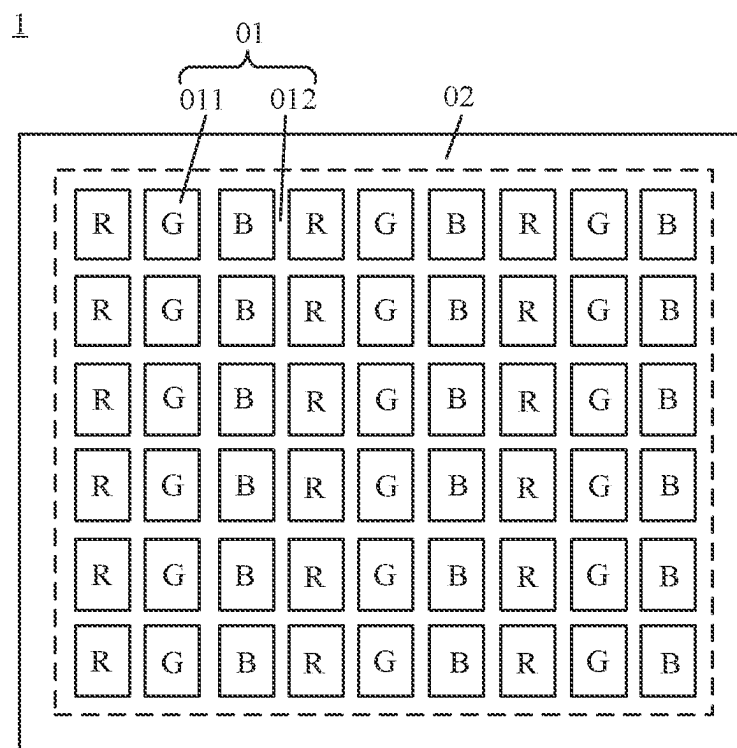


FIG. 2A

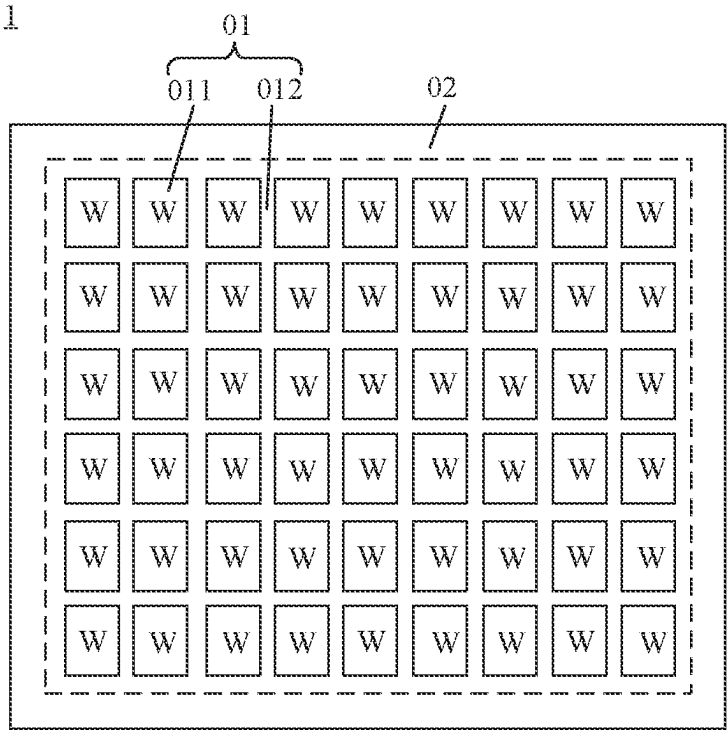


FIG. 2B

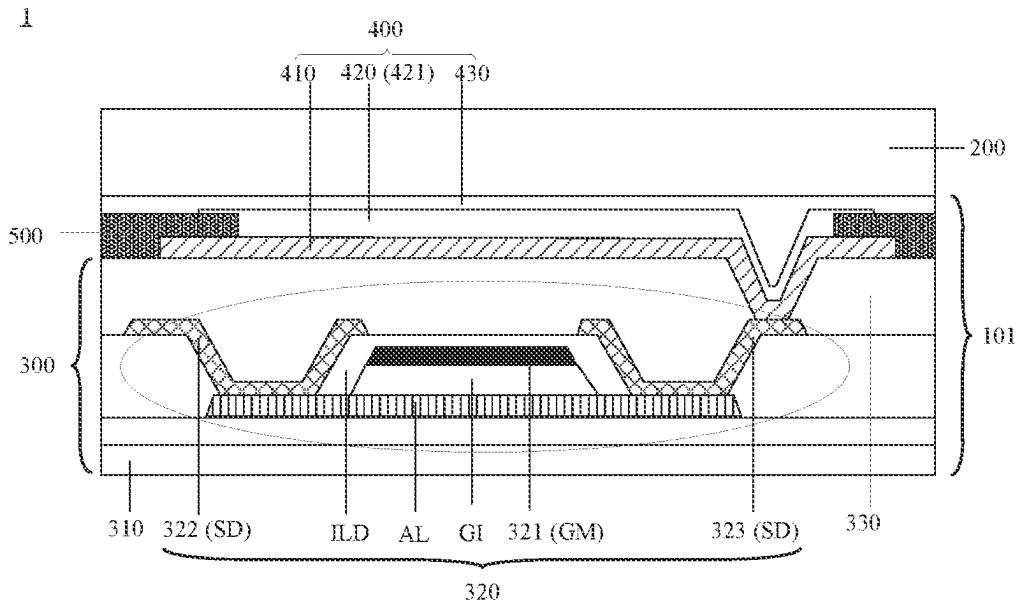


FIG. 3

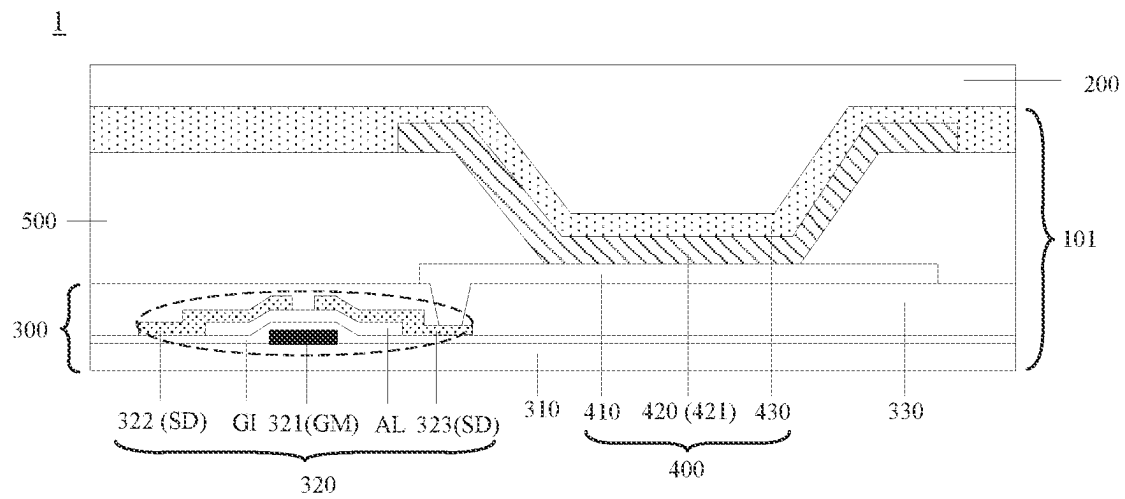


FIG. 4

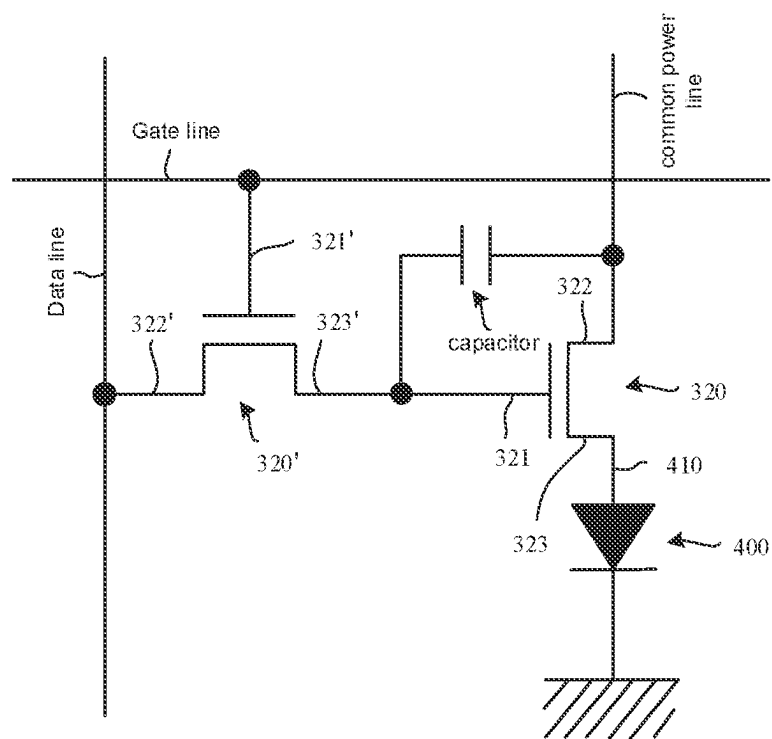


FIG. 5

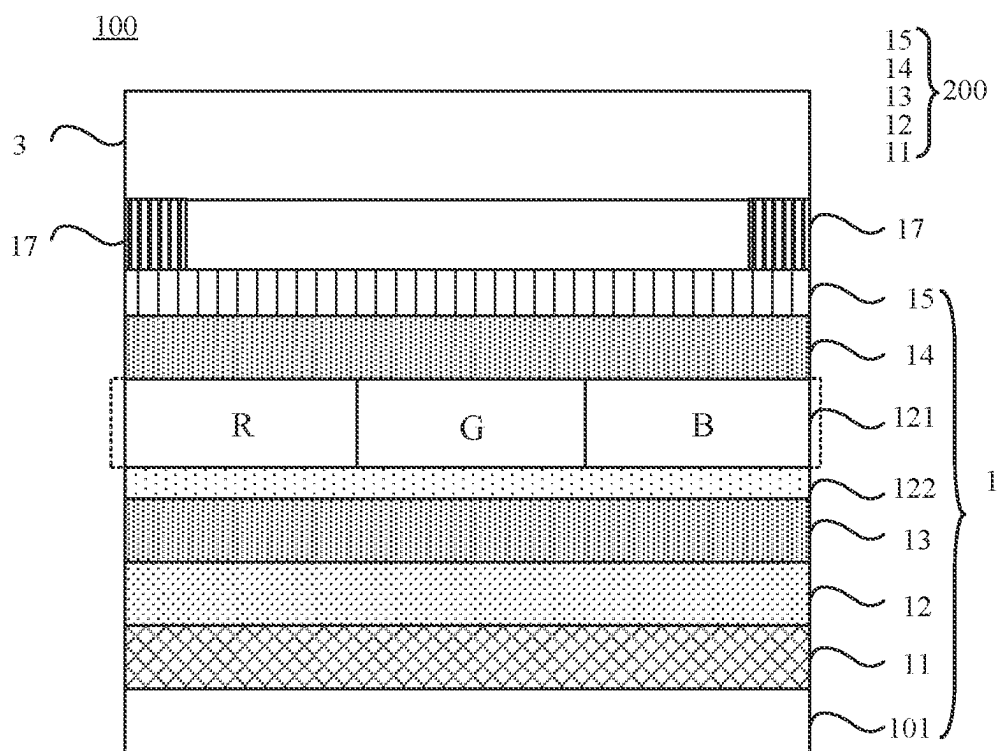


FIG. 6

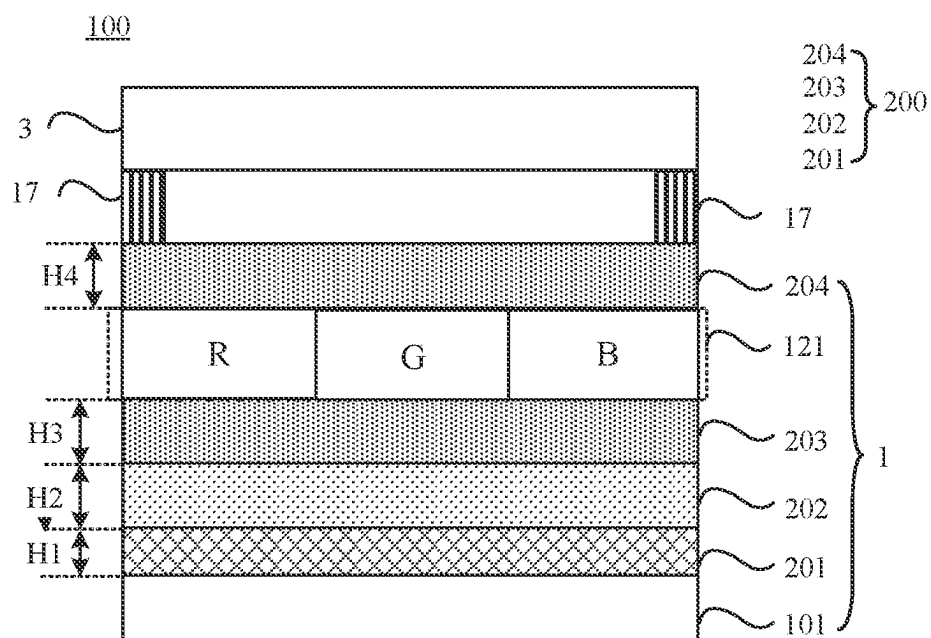


FIG. 7

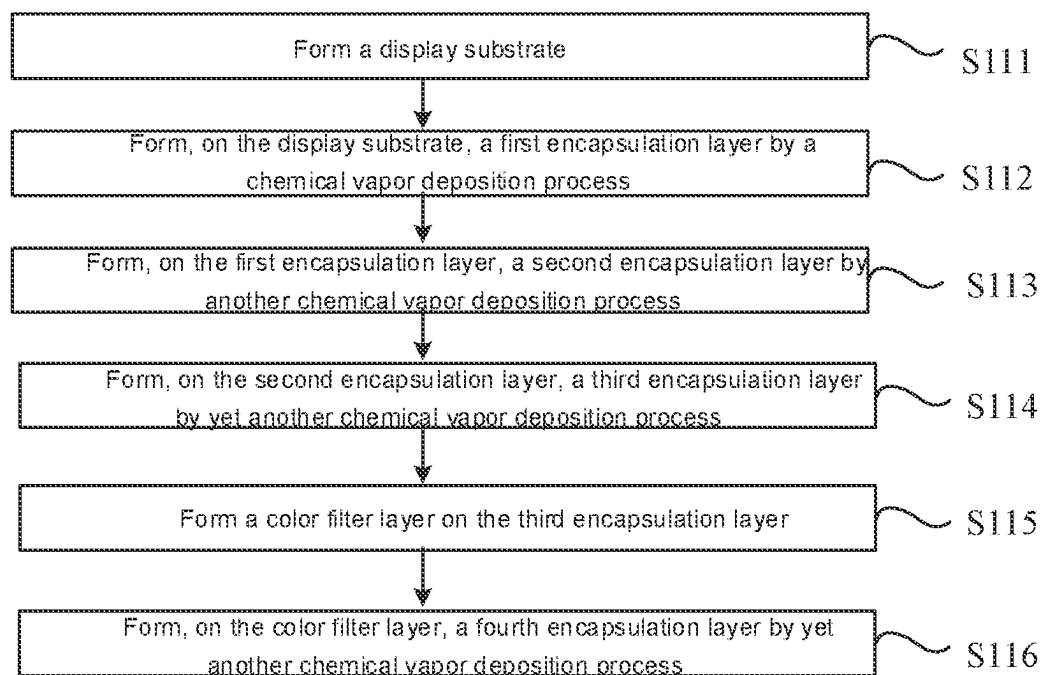


FIG. 8

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**DISPLAY PANEL WITH CVD
ENCAPSULATION LAYERS, DISPLAY
APPARATUS COMPRISING THE
ENCAPSULATION LAYERS, AND METHOD
FOR MANUFACTURING THE DISPLAY
PANEL**

CROSS-REFERENCE TO RELATED
APPLICATIONS

This application is a national phase entry under 35 USC 371 of International Patent Application No. PCT/CN2021/089381, filed on Apr. 23, 2021, which claims priority to Chinese Patent Application No. 202010328116.3, filed on Apr. 23, 2020, which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

The present disclosure relates to the field of display technologies, and in particular, to a display panel, a display apparatus and a method for manufacturing the display panel.

BACKGROUND

OLED (organic light-emitting diode) display apparatuses have become a mainstream development trend of display apparatuses at present due to their advantages such as thinness and lightness, fast response, wide viewing angle, flexible folding and high contrast. OLED display apparatuses are very sensitive to water vapor and oxygen. In a case where the water vapor and oxygen enter the OLED display apparatus, organic materials of the OLED display apparatus are prone to decay, and thus a service life of the OLED display apparatus is shortened. Therefore, in order to prevent the entering of the water vapor and oxygen, it is necessary to encapsulate the OLED display apparatus, so as to prevent the entering of the water vapor and oxygen, thereby avoiding aging of the organic materials. As a result, the service life of the display apparatus is prolonged.

SUMMARY

In an aspect, a display panel is provided. The display panel includes a display substrate and an encapsulation layer. The display substrate includes: a base substrate; and a plurality of light-emitting devices disposed on the base substrate, the plurality of light-emitting devices being configured to emit light. The encapsulation layer is located on a side of the light-emitting devices away from the base substrate, and is configured to encapsulate the light-emitting devices. The encapsulation layer includes: a second encapsulation layer located on the side of the light-emitting devices away from the base substrate, and made of a silicon nitride material; and a third encapsulation layer located on a side of the second encapsulation layer away from the light-emitting devices, and made of a silicon carbonitride material or a silicon carbide material.

In some embodiments, the encapsulation layer further includes a first encapsulation layer located between the second encapsulation layer and the light-emitting devices, and made of a silicon oxynitride material.

In some embodiments, the first encapsulation layer is a first encapsulation layer manufactured by a chemical vapor deposition process; the second encapsulation layer is a second encapsulation layer manufactured by another chemical vapor deposition process; and the third encapsulation

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layer is a third encapsulation layer manufactured by yet another chemical vapor deposition process.

In some embodiments, the encapsulation layer further includes a fourth encapsulation layer located on a side of the third encapsulation layer away from the second encapsulation layer, and made of a silicon carbonitride material or a silicon carbide material.

In some embodiments, the first encapsulation layer is a first encapsulation layer manufactured by a chemical vapor deposition process; the second encapsulation layer is a second encapsulation layer manufactured by another chemical vapor deposition process; the third encapsulation layer is a third encapsulation layer manufactured by yet another chemical vapor deposition process; and the fourth encapsulation layer is a fourth encapsulation layer manufactured by yet another chemical vapor deposition process.

In some embodiments, the display panel further includes a color filter layer located on a side of the third encapsulation layer away from the base substrate. The third encapsulation layer is made of the silicon carbide material.

In some embodiments, the display panel further includes a fourth encapsulation layer located on a side of the color filter layer away from the third encapsulation layer. The fourth encapsulation layer is made of a silicon carbonitride material.

In some embodiments, the base substrate is a silicon-based backplane integrated with a plurality of pixel circuits. Each of the light-emitting devices includes a first electrode, a light-emitting functional layer and a second electrode. The plurality of pixel circuits are electrically connected to respective first electrodes of the light-emitting devices.

In some embodiments, the display panel further includes a color filter layer located on a side of the third encapsulation layer away from the base substrate, the third encapsulation layer being made of the silicon carbide material. The encapsulation layer further includes: a first encapsulation layer located between the second encapsulation layer and the light-emitting devices, and made of a silicon oxynitride material; and a fourth encapsulation layer located on a side of the color filter layer away from the third encapsulation layer, and made of a silicon carbonitride material. The base substrate is a silicon-based backplane integrated with a plurality of pixel circuits. Each of the light-emitting devices includes a first electrode, a light-emitting functional layer and a second electrode. The plurality of pixel circuits are electrically connected to respective first electrodes of the light-emitting devices.

In some embodiments, a thickness of the first encapsulation layer is in a range of 1000 Å to 10000 Å, inclusive; a thickness of the second encapsulation layer is in a range of 1000 Å to 10000 Å, inclusive; a thickness of the third encapsulation layer is in a range of 2000 Å to 20000 Å, inclusive; and a thickness of the fourth encapsulation layer is in a range of 1000 Å to 10000 Å, inclusive.

In another aspect, a display panel is provided. The display panel includes a display substrate and an encapsulation layer. The display substrate includes: a base substrate; and a plurality of light-emitting devices disposed on the base substrate, the plurality of light-emitting devices being configured to emit light. The encapsulation layer includes a first inorganic layer, a second inorganic layer, and a first organic layer that are stacked on the display substrate in sequence. The first inorganic layer is made of a silicon nitride material; the second inorganic layer is made of a silicon oxide material; and the first organic layer is made of a parylene material.

In some embodiments, the first inorganic layer is a first inorganic layer manufactured by a chemical vapor deposition process; the second inorganic layer is a second inorganic layer manufactured by an atomic layer deposition process; and the first organic layer is a first organic layer manufactured by a molecular layer deposition process.

In some embodiments, the display panel further includes a color filter layer located on a side of the first organic layer away from the second inorganic layer. The encapsulation layer further includes: a second organic layer located on a side of the color filter layer away from the first organic layer, and made of a parylene material; and a third inorganic layer located on a side of the second organic layer away from the first organic layer, and made of a silicon oxide material.

In some embodiments, the second organic layer is a second organic layer manufactured by a molecular layer deposition process; and the third inorganic layer is a third inorganic layer manufactured by an atomic layer deposition process.

In some embodiments, the base substrate is a silicon-based backplane integrated with a plurality of pixel circuits. Each of the light-emitting devices includes a first electrode, a light-emitting functional layer and a second electrode. The plurality of pixel circuits are electrically connected to respective first electrodes of the light-emitting devices.

In yet another aspect, a display apparatus is provided. The display apparatus includes the display panel of any one of the above, and a glass cover plate configured to protect the display panel.

In yet another aspect, a method for manufacturing a display panel is provided. The method includes: forming a display substrate; forming, on the display substrate, a second encapsulation layer by a chemical vapor deposition process; and forming, on the second encapsulation layer, a third encapsulation layer by another chemical vapor deposition process.

In some embodiments, the first encapsulation layer is made of a silicon oxynitride material; the second encapsulation layer is made of a silicon nitride material; the third encapsulation layer is made of a silicon carbide material; and the fourth encapsulation layer is made of a silicon carbonitride material.

In some embodiments, before forming, on the display substrate, the second encapsulation layer by the chemical vapor deposition process, the method further includes: forming, on the display substrate, a first encapsulation layer by yet another chemical vapor deposition process; or after forming, on the second encapsulation layer, the third encapsulation layer by the another chemical vapor deposition process, the method further includes: forming a color filter layer on the third encapsulation layer; and forming, on the color filter layer, a fourth encapsulation layer by yet another chemical vapor deposition process; or before forming, on the display substrate, the second encapsulation layer by the chemical vapor deposition process, the method further comprises: forming, on the display substrate, a first encapsulation layer by yet another chemical vapor deposition process; and after forming, on the second encapsulation layer, the third encapsulation layer by the another chemical vapor deposition process, the method further includes: forming a color filter layer on the third encapsulation layer; and forming, on the color filter layer, a fourth encapsulation layer by yet another chemical vapor deposition process.

In some embodiments, forming the display substrate includes: forming a silicon-based backplane integrated with a plurality of pixel circuits.

BRIEF DESCRIPTION OF THE DRAWINGS

In order to describe technical solutions in the present disclosure more clearly, accompanying drawings to be used in some embodiments of the present disclosure will be introduced briefly below. However, the accompanying drawings to be described below are merely accompanying drawings of some embodiments of the present disclosure, and a person of ordinary skill in the art can obtain other drawings according to these drawings. In addition, the accompanying drawings in the following description may be regarded as schematic diagrams, and are not limitations on actual sizes of products, actual processes of methods and actual timings of signals involved in the embodiments of the present disclosure.

FIG. 1 is a structural diagram of a display apparatus, in accordance with some embodiments;

FIG. 2A is a diagram of divided regions of a display panel, in accordance with some embodiments;

FIG. 2B is another diagram of divided regions of a display panel, in accordance with some embodiments;

FIG. 3 is a structural diagram of a display panel, in accordance with some embodiments;

FIG. 4 is a structural diagram of another display panel, in accordance with some embodiments;

FIG. 5 is an equivalent circuit diagram of a pixel circuit, in accordance with some embodiments;

FIG. 6 is a structural diagram showing a display panel and a glass cover plate, in accordance with some embodiments;

FIG. 7 is another structural diagram showing a display panel and a glass cover plate, in accordance with some embodiments; and

FIG. 8 is a flowchart of a method for manufacturing a display panel, in accordance with some embodiments.

DETAILED DESCRIPTION

Technical solutions in some embodiments of the present disclosure will be described clearly and completely below with reference to the accompanying drawings. However, the described embodiments are merely some but not all embodiments of the present disclosure. All other embodiments obtained by a person of ordinary skill in the art based on the embodiments of the present disclosure shall be included in the protection scope of the present disclosure.

Unless the context requires otherwise, throughout the description and the claims, the term “comprise” and other forms thereof such as the third-person singular form “comprises” and the present participle form “comprising” are construed as an open and inclusive meaning, i.e., “including, but not limited to”. In the description of the specification, the terms such as “one embodiment”, “some embodiments”, “exemplary embodiments”, “example”, “specific example” and “some examples” are intended to indicate that specific features, structures or characteristics related to the embodiment(s) or example(s) are included in at least one embodiment or example of the present disclosure. Schematic representations of the above terms do not necessarily refer to the same embodiment(s) or examples(s). In addition, the specific features, structures, materials or characteristics may be included in any one or more embodiments or examples in any suitable manner.

Hereinafter, the terms “first” and “second” are only used for descriptive purposes, and are not to be construed as indicating or implying the relative importance or implicitly indicating the number of indicated technical features. Thus, a feature defined with the term “first” or “second” may

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explicitly or implicitly include one or more of the features. In the description of the embodiments of the present disclosure, the terms “a plurality of”, “the plurality of” and “multiple” each mean two or more unless otherwise specified.

In the description of some embodiments, terms such as “coupled” and “connected” and derivatives thereof may be used. For example, the term “connected” may be used in the description of some embodiments to indicate that two or more components are in direct physical or electrical contact with each other. As another example, the term “coupled” may be used in the description of some embodiments to indicate that two or more components are in direct physical or electrical contact. However, the term “coupled” or “communicatively coupled” may also mean that two or more components are not in direct contact with each other, but still cooperate or interact with each other. The embodiments disclosed herein are not necessarily limited to the content herein.

The phrase “at least one of A, B and C” has the same meaning as the phrase “at least one of A, B or C”, and they both include the following combinations of A, B and C: only A, only B, only C, a combination of A and B, a combination of A and C, a combination of B and C, and a combination of A, B and C. The phrase “A and/or B” includes the following three combinations: only A, only B, and a combination of A and B.

As used herein, depending on the context, the term “if” is optionally construed as “when”, “in a case where”, “in response to determining” or “in response to detecting”. Similarly, depending on the context, the phrase “if it is determined” or “if [a stated condition or event] is detected” is optionally construed as “in a case where it is determined”, “in response to determining”, “in a case where [the stated condition or event] is detected”, or “in response to detecting [the stated condition or event]”.

The phrase “applicable to” or “configured to” used herein has an open and inclusive meaning, which does not exclude apparatuses that are applicable to or configured to perform additional tasks or steps.

In addition, the phrase “based on” is meant to be open and inclusive, since a process, step, calculation or other action that is “based on” one or more of the stated conditions or values may, in practice, be based on additional conditions or values other than those stated.

As used herein, terms such as “about”, “substantially” or “approximately” include a stated value and an average value within an acceptable range of deviation of a particular value. The acceptable range of deviation is determined by a person of ordinary skill in the art in view of the measurement in question and errors associated with the measurement of a particular quantity (i.e., limitations of the measurement system).

Exemplary embodiments are described herein with reference to sectional views and/or plan views as idealized exemplary drawings. In the accompanying drawings, thicknesses of layers and sizes of regions are enlarged for clarity. Thus, variations in shapes relative to the accompanying drawings due to, for example, manufacturing technologies and/or tolerances may be envisaged. Therefore, the exemplary embodiments should not be construed as being limited to the shapes of the regions shown herein, but including shape deviations due to, for example, manufacturing. For example, an etched region shown to have a rectangular shape generally has a feature of being curved. Therefore, the regions shown in the accompanying drawings are schematic in nature, and their shapes are not intended to show actual

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shapes of the regions in an apparatus, and are not intended to limit the scope of the exemplary embodiments.

In some embodiments of the present disclosure, as shown in FIG. 1, a display apparatus 100 is provided. The display apparatus 100 may include a display panel 1, a frame 2, a glass cover plate 3, a circuit board 4 and other electronic accessories.

As shown in FIG. 1, a longitudinal section of the frame 2 is U-shaped, the display panel 1, the circuit board 4 and the other electronic accessories are all disposed in the frame 2, the circuit board 4 is disposed behind the display panel 1, and the glass cover plate 3 is disposed on a side of the display panel 1 away from the circuit board 4.

The display apparatus 100 may be an organic light-emitting diode (OLED) display apparatus. In this case, the display panel 1 is an organic electroluminescent display panel. The display apparatus 100 may also be a quantum dot light-emitting diode (QLED) display apparatus. In this case, the display panel 1 is a quantum dot electroluminescent display panel. For convenience of description, the following embodiments are all described by taking an example in which the display apparatus 100 is the OLED display apparatus.

As shown in FIG. 2A, it can be found that the display panel 1 has a display area 01 and a peripheral area 02 located on at least one side of the display area 01 when viewed from a viewing angle directly facing a display surface of the display panel 1. FIGS. 2A and 2B illustrate an example in which the peripheral area 02 surrounds the display area 01, which is not limited in the embodiments of the present disclosure.

With reference to FIG. 2A, the display area 01 may include a plurality of light-emitting regions 011 (each light-emitting region 011 corresponding to a sub-pixel) and a non-light-emitting region 012. In some embodiments, the plurality of light-emitting regions 011 may include red light-emitting regions for emitting red light, blue light-emitting regions for emitting blue light, and green light-emitting regions for emitting green light. In some embodiments, as shown in FIG. 2A, a column of red light-emitting regions R, a column of green light-emitting regions G and a column of blue light-emitting regions B are alternately arranged in turn. In some other embodiments, an arrangement order of the red light-emitting regions R, the green light-emitting regions G, and the blue light-emitting regions B may be changed.

In some other embodiments, the plurality of light-emitting regions 011 may all be used for emitting white light. For example, as shown in FIG. 2B, each light-emitting region 011 is a white light-emitting region W for emitting white light.

In addition, the peripheral area 02 in FIG. 2A or 2B is used for wiring, so that a plurality of sub-pixels in the display area 01 are connected to gate drive circuit(s) or source drive circuit(s) through the wiring. Moreover, the gate drive circuit(s) may also be arranged in the peripheral area 02 by using the gate driver on array (GOA) technology, and thus a size of the peripheral area 02 is reduced. As a result, a narrow bezel of the display apparatus is realized.

In order to further describe the display panel 1 in some embodiments of the present disclosure, an internal structure of the display panel 1 is described in detail below. It will be noted that, since a structure corresponding to each light-emitting region 011 of the display panel 1 is basically repeated, FIGS. 3 and 4 will be illustrated below by taking an example in which only a portion of the structure corre-

sponding to the light-emitting region **011** of the display panel **1** in FIG. 2A or 2B is showed.

In some embodiments of the present disclosure, as shown in FIG. 3, the display panel **1** may include a display substrate **101** and an encapsulation layer **200** for encapsulating the display substrate **101**. The encapsulation layer **200** may be used to prevent water vapor and oxygen from entering the display substrate **101**.

Firstly, a structure of the display substrate **101** will be exemplarily described below. As shown in FIG. 3, the display substrate **101** may include a base substrate **300** and light-emitting devices **400** disposed on the base substrate **300**.

It is easily understood that the base substrate **300** may be a blank substrate, or a substrate with film structure(s).

In some embodiments, the base substrate **300** includes a blank substrate **310**, and pixel circuits disposed on the blank substrate **310**.

In some embodiments, the blank substrate **310** may be a flexible blank substrate or a rigid blank substrate. A material of the flexible blank substrate **310** may be, for example, polyimide (PI). A material of the rigid blank substrate **310** may be, for example, glass.

In some other embodiments, the base substrate **300** may be a silicon-based backplane. It will be noted that an accuracy of pixel circuits included in the silicon-based backplane is much higher than an accuracy of pixel circuits manufactured on a glass substrate or a polyimide substrate. For example, a size of the pixel circuits included in the silicon-based backplane may reach a level required by a chip for a size of a circuit device. However, an accuracy that can be achieved by current manufacturing processes of the light-emitting devices **400** is difficult to match the accuracy of the pixel circuits included in the silicon-based backplane. As a result, in the case where the base substrate **300** is the silicon-based backplane, the plurality of light-emitting regions **011** are generally white light-emitting regions for emitting white light (as shown in FIG. 2B).

The base substrate **300** further includes gate lines arranged in a direction on the blank substrate **310**, and data lines and common power lines that are insulated from and cross the gate lines on the blank substrate **310**. The common power lines are generally parallel to the data lines. The plurality of sub-pixels may be defined by the gate lines and the data lines (or both the data lines and the common power lines) that are arranged crosswise. Each sub-pixel has a pixel circuit, and the pixel circuit is electrically connected to a light-emitting device **400** for driving the light-emitting device **400** to emit light.

The pixel circuit may include at least one thin film transistor (abbreviated as: TFT) **320** and at least one capacitor (not shown in FIG. 3). Each thin film transistor **320** may be of a top-gate structure or a bottom-gate structure. As shown in FIG. 3, in the case where the thin film transistor **320** has the top-gate structure, the thin film transistor **320** includes an active layer AL, a gate insulating layer GI, a gate metal layer GM (i.e., a gate electrode **321**), an interlayer dielectric layer ILD, and a source-drain metal layer SD (including a source electrode **322** and a drain electrode **323**) that are disposed on the blank substrate **310** in sequence. In some other embodiments, as shown in FIG. 4, in the case where the thin film transistor **320** has the bottom-gate structure, the thin film transistor **320** includes a gate metal layer GM (i.e., a gate electrode **321**), a gate insulating layer GI, an active layer AL, and a source-drain metal layer SD (including a source electrode **322** and a drain electrode **323**) that are disposed on the blank substrate **310** in sequence.

The active layer AL of the thin film transistor **320** may be made of amorphous silicon, monocrystalline silicon, polycrystalline silicon, or an oxide semiconductor. The active layer AL includes a channel region that is not doped with any impurity, and a source region and a drain region that are located on two sides of the channel region and formed by being doped with an impurity. The doped impurity changes as the type of the thin film transistor **320** changes, and may be an N-type impurity or a P-type impurity.

The capacitor includes a first electrode plate and a second electrode plate. An interlayer insulating film is provided between the first electrode plate and the second electrode plate as a dielectric.

Electrical connection relationships of an inside and an outside of the pixel circuit will be described below by taking an example in which the pixel circuit includes two thin film transistors (i.e., a switching thin film transistor and a driving thin film transistor) and one capacitor, which constitutes a 2T1C structure. Although FIG. 3 only shows structures of the driving thin film transistor **320** (e.g., the structure in the dashed circle in FIG. 3) and the light-emitting device **400** and a connection relationship therebetween, according to the description of the context, those skilled in the art may fully determine the structure of the switching transistor and connection relationships between the switching transistor and other components.

As shown in FIG. 5, a gate **321'** of the switching thin film transistor **320'** is connected to a gate line, a source **322'** of the switching thin film transistor **320'** is connected to a data line, and a drain **323'** of the switching thin film transistor **320'** is connected to a gate **321** of the driving thin film transistor **320**. A source **322** of the driving thin film transistor **320** is connected to a common power line, and a drain **323** of the driving thin film transistor **320** is connected to a first electrode **410** of the light-emitting device **400** through a via hole. The first electrode plate of the capacitor is connected to the gate **321** of the driving thin film transistor **320**, and the second electrode plate of the capacitor is connected to the source **322** of the driving thin film transistor **320**.

The switching thin film transistor **320'** is turned on by a gate voltage applied to the gate line, so that a data voltage applied to the data line is transmitted to the driving thin film transistor **320**. There is a certain difference between the data voltage transmitted from the switching thin film transistor **320'** to the driving thin film transistor **320** and the common voltage applied to the driving thin film transistor **320** from the common power line. A voltage equivalent to an absolute value of the difference is stored in the capacitor, and a current corresponding to the voltage stored in the capacitor flows into the light-emitting device **400** through the driving thin film transistor **320**, so that the light-emitting device **400** emits light.

In addition, as shown in FIGS. 3 and 4, the light-emitting device **400** includes a first electrode **410**, a light-emitting functional layer **420**, and a second electrode **430** that are disposed on the base substrate **300** in sequence. One of the first electrode **410** and the second electrode **430** is an anode (for providing holes), and the other one of the first electrode **410** and the second electrode **430** is a cathode (for providing electrons). The first electrode **410** and the second electrode **430** respectively inject the holes and the electrons into the light-emitting functional layer **420**, and the light-emitting functional layer **122** emits light when excitons generated by the combination of the holes and the electrons transitions from an excited state to a ground state.

The first electrode **410** may be made of a metal with high reflectivity, and the second electrode **430** may be made of a transparent conductive material. In this case, light from the light-emitting functional layer **420** is reflected by the first electrode **410** and exit through the second electrode **430**. Thus, the light-emitting device **400** is a top-emission light-emitting device. However, it is not limited thereto. In a case where the first electrode **410** is made of a transparent conductive material and the second electrode **430** is material of a metal with high reflectivity, a bottom-emission light-emitting device may be formed. Of course, in a case where the first electrode **410** and the second electrode **430** each are made of a transparent conductive material, a double-sided-emission light-emitting device may be formed.

The transparent conductive material may be, for example, indium tin oxide (ITO), indium zinc oxide (IZO) or indium gallium zinc oxide (IGZO). The metal with high reflectivity may be, for example, Ag.

In some embodiments, as shown in FIGS. **3** and **4**, the light-emitting functional layer **420** may include a light-emitting layer **421**. In some other embodiments, in addition to the light-emitting layer **421**, the light-emitting functional layer **420** may further include at least one of a hole injection layer (HIL), a hole transport layer (HTL), an electron transport layer (ETL) and an electron injection layer (EIL) (not shown in FIGS. **3** and **4**). In a case where the light-emitting functional layer **420** includes all the above layers, the hole injection layer, the hole transport layer, the light-emitting layer **421**, the electron transport layer, and the electron injection layer are sequentially stacked on the first electrode **410** as the anode.

In some embodiments, as shown in FIGS. **3** and **4**, the base substrate **300** may further include a planarization layer **330** disposed between thin film transistors **320** and first electrodes **410**, and a pixel define layer **500** disposed on a side of the first electrodes **410** away from the blank substrate **310**. The pixel define layer **500** has a plurality of openings and a barrier wall disposed around each opening. The light-emitting device **400** is disposed in an opening region. First electrodes **410** and light-emitting layers **421** of adjacent light-emitting devices **400** are separated by the barrier wall of the pixel define layer **500**.

In the case where the light-emitting device **400** includes the at least one of the hole injection layer, the hole transport layer, the electron transport layer and the electron injection layer, the at least one of the hole injection layer, the hole transport layer, the electron transport layer and the electron injection layer of the light-emitting device **400** may be disconnected from that of adjacent light-emitting devices **400**. That is, the at least one of the hole injection layer, the hole transport layer, the electron transport layer and the electron injection layer is only arranged in the opening. Alternatively, the at least one of the hole injection layer, the hole transport layer, the electron transport layer and the electron injection layer of the light-emitting device **400** may be formed a whole layer together with that of other light-emitting devices **400**.

Since the light-emitting functional layer **420** is very sensitive to the water vapor and oxygen in the air, once the light-emitting functional layer **420** contacts the water vapor and oxygen in the air, organic materials of the light-emitting functional layer **420** may chemically react with the water vapor and oxygen. As a result, a performance of the OLED display apparatus is degraded and a service life of the OLED display apparatus is shortened.

In order to ameliorate the above technical problems, it can be seen from the above, as shown in FIGS. **3** and **4**, the

display panel **1** may further include an encapsulation layer **200** for encapsulating the display substrate **101**. The encapsulation layer **200** may be used to prevent the water vapor and oxygen from entering the display substrate **101**.

Some embodiments of the present disclosure provide different structures of the encapsulation layer **200** for the display substrate **101**. The different structures of the encapsulation layers **200** will be described in detail below. It will be noted that, in order to simplify the accompany drawings, the following accompany drawings do not show the structure of the display substrate **101**.

As described above, the base substrate **300** in the display substrate **101** may be the glass substrate, the polyimide substrate, or the silicon-based backplane integrated with the pixel circuits. The accuracy of the pixel circuits included in the silicon-based backplane is much higher than the accuracy of the pixel circuits manufactured on the glass substrate or the polyimide substrate. With a rapid growth of market demand, especially in recent years, with a significant growth of the market demand for virtual reality (VR) and augmented reality (AR) technologies, display panels **1** are moving towards miniaturization and high pixel density at an accelerated pace. The display panels **1** adopting the silicon-based backplanes are becoming a focus of attention in the display field by virtue of their advantages of miniaturization and high pixel density.

In the case where the silicon-based backplane is used as the base substrate **300**, since the accuracy that can be achieved by the manufacturing processes of the light-emitting devices **400** is difficult to match the accuracy of the pixel circuits included in the silicon-based backplane, the plurality of light-emitting regions **011** are generally the white light-emitting regions for emitting white light (as shown in FIG. **2B**). In this case, a color filter layer needs to be provided on the display substrate **101**. The color filter layer includes three types of photoresists (i.e., RGB photoresists) corresponding to three types of sub-pixels (i.e., RGB sub-pixels), so as to filter the white light emitted by the light-emitting devices **400** to convert the white light into light of three primary colors (i.e., light of RGB colors).

It will be understood that, in the case where the silicon-based backplane is used as the base substrate **300**, the plurality of light-emitting regions **011** may still include the red light-emitting regions R, the green light-emitting regions G, and the blue light-emitting regions B (as shown in FIG. **2A**). For example, in a case where the accuracy that can be achieved by manufacturing processes of the light-emitting devices **400** may match the accuracy of the pixel circuits included in the silicon-based backplane, or users have less requirements for the pixel density of the display panel **1**, a plurality of light-emitting devices **400** corresponding to the plurality of light-emitting regions **011** may include light-emitting devices **400** for emitting red light, light-emitting devices **400** for emitting green light, and light-emitting devices **400** for emitting blue light.

For convenience of description, unless otherwise specified, the following embodiments will be described by taking an example in which the plurality of light-emitting devices **400** are configured to emit white light.

As shown in FIG. **6**, the encapsulation layer **200** includes a first inorganic layer **11**, a second inorganic layer **12**, a first organic layer **13**, a second organic layer **14** and a third inorganic layer **15** that are disposed on the display substrate **101** in sequence. Moreover, as shown in FIG. **6**, the display panel **1** further includes a color filter layer **121** located between the first organic layer **13** and the second organic layer **14**. The glass cover plate **3** for covering the display

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panel **1** is located on a side of the third inorganic layer **15** away from the display substrate **101**, and the glass cover plate **3** is bonded to the display panel **1** through a frame sealant **17**.

The color filter layer **121** is generally manufactured by a photolithography process. It will be noted that, the photolithography process typically refers to a process of forming a specific pattern by using a mask and removing photoresist. Forming the specific pattern includes coating a surface of a film with photoresist; exposing the photoresist; developing after exposing the photoresist; and etching regions of the film exposed by the photoresist, so as to form a required pattern.

The first inorganic layer **11** is made of a silicon nitride (SiNx) material, and may be manufactured by a chemical vapor deposition (CVD) equipment. Since the first inorganic layer **11** made of the silicon nitride material has good denseness and intactness, it may be possible to prevent the entering of the water vapor and oxygen. Therefore, by providing the first inorganic layer **11** in the encapsulation layer **200**, it may be possible to prevent the water vapor and oxygen from entering the display substrate **101**, so as to avoid the aging of the light-emitting functional layer **420**. As a result, the service life of the display apparatus **100** is prolonged.

The second inorganic layer **12** is made of a silicon oxide material, and may be manufactured by an atomic layer deposition (ALD) equipment. The second inorganic layer **12** may play a role of blocking the water vapor and oxygen for the display substrate **101** located below the second inorganic layer **12**.

The first organic layer **13** is made of a parylene material, and may be manufactured by a molecular layer deposition (MLD) equipment. The first organic layer **13** may play a role of planarizing the display substrate **101** located below the first organic layer **13** and wrapping particles of the display substrate **101**.

The color filter layer **121** includes three types of photoresists (i.e., RGB photoresists). In a manufacturing process of the color filter layer **121**, a process procedure and process parameters of the manufacturing process of the color filter layer **121** are particularly important, and fineness of the manufacturing process of the color filter layer **121** determines a resolution of a silicon-based display panel. Low temperature color filter technology is a color filter technology applied to the silicon-based display panel, which can achieve pixel processing accuracy of 1-2 microns. A problem of the low temperature color filter technology is that a low temperature curing process for curing the three photoresist materials (i.e., RGB photoresist materials) results in poor adhesion between the color filter layer **121** and the first organic layer **13**. In order to ameliorate the technical problem, as shown in FIG. 6, the display panel **1** further includes an adhesive layer **122** disposed between the color filter layer **121** and the first organic layer **13**. By providing the adhesive layer **122** in the display panel **1**, it is possible to improve an adhesive force between the color filter layer **121** and the first organic layer **13**, which improves a reliability of the display apparatus **100**.

The second organic layer **14** is made of a parylene material and is manufactured by the MLD equipment. The second organic layer **14** may play a role of planarizing the color filter layer **121** located below the second organic layer **14** and wrapping particles of the color filter layer **121**.

The third inorganic layer **15** is made of a silicon oxide (SiO₂) material and is manufactured by the ALD equipment. By providing the third inorganic layer **15** in the display panel

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1, and utilizing an inorganic property of the third inorganic layer **15**, the display panel **1** may be better bonded to the glass cover plate **3**, which prolongs the service life of the display apparatus **100**.

Thus, by manufacturing the encapsulation layer **200** on the display substrate **101**, it is possible to effectively prevent the water vapor and oxygen from entering the display substrate **101**, so as to avoid the aging of the light-emitting functional layer **420**. As a result, the service life of the display apparatus **100** is prolonged. For example, the encapsulation layer **200** includes the first inorganic layer **11**, the second inorganic layer **12**, the first organic layer **13**, the second organic layer **14** and the third inorganic layer **15** that are disposed on the display substrate **101** in sequence.

It will be noted that, in some embodiments, in a case where requirements for the service life and performance of the display apparatus **100** are not high, the first inorganic layer **11** in the encapsulation layer **200** may be omitted. In this way, the process may be simplified, and thus the cost may be reduced. In addition, a thickness of the display apparatus **100** may be reduced, and thus a thin and light design of the display apparatus **100** may be realized. In this case, the encapsulation layer **200** composed of the second inorganic layer **12**, the first organic layer **13**, the second organic layer **14** and the third inorganic layer **15** may also prevent the water vapor and oxygen from entering the display substrate **101**.

In some embodiments, in the case where the plurality of light-emitting devices **400** includes the light-emitting devices **400** for emitting red light, the light-emitting devices **400** for emitting green light, and the light-emitting devices **400** for emitting blue light, there is no need to provide the color filter layer **121** in the display panel **1**. In this case, the second organic layer **14** may also be omitted. In this way, the process is simplified, and thus the cost is reduced. In this case, the encapsulation layer **200** composed of the first inorganic layer **11**, the second inorganic layer **12**, the first organic layer **13** and the third inorganic layer **15** may prevent the water vapor and oxygen from entering the display substrate **101**.

In some embodiments, the third inorganic layer **15** may also be omitted.

In some embodiments, at least two of the first inorganic layer **11**, the color filter layer **121**, the second organic layer **14** and the third inorganic layer **15** may be omitted. In this way, the process is further simplified, and thus the cost is further reduced. Moreover, the thickness of the display apparatus **100** is further reduced, thereby realizing a thin and light design of the display apparatus **100**.

In some embodiments, as shown in FIG. 7, another encapsulation layer **200** is provided. As shown in FIG. 7, the encapsulation layer **200** includes a first encapsulation layer **201**, a second encapsulation layer **202**, a third encapsulation layer **203** and a fourth encapsulation layer **204** that are disposed on the display substrate **101** in sequence. The first encapsulation layer **201**, the second encapsulation layer **202**, the third encapsulation layer **203** and the fourth encapsulation layer **204** each are manufactured by a chemical vapor deposition process. As shown in FIG. 7, the display panel **1** further includes a color filter layer **121** located between the third encapsulation layer **203** and the fourth encapsulation layer **204**. The glass cover plate **3** for covering the display panel **1** is located on a side of the fourth encapsulation layer **204** away from the display substrate **101**, and the glass cover plate **3** is bonded to the display panel **1** by a sealant **17**.

The first encapsulation layer **201** may be made of a silicon oxynitride (SiON) material. The silicon oxynitride (SiON)

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contains an oxygen element, so that a surface of the first encapsulation layer **201** may have a good interface quality, and have good water resistance. In addition, a refractive index of the silicon oxynitride is small (e.g., the refractive index of the silicon oxynitride is smaller than a refractive index of silicon nitride), which helps adjust an antireflection effect of laminated films of thin film encapsulation (TFE), so as to increase light extraction efficiency of the light-emitting devices **400**.

The embodiments of the present disclosure do not limit a specific value of a thickness of the first encapsulation layer **201**. As shown in FIG. 7, the thickness H1 of the first encapsulation layer **201** may be in a range of 1000 Å to 10000 Å, inclusive (i.e., $1000 \text{ Å} \leq H1 \leq 10000 \text{ Å}$). For example, the thickness H1 of the first encapsulation layer **201** may be 1500 Å, 3500 Å, 5000 Å, 6500 Å, or 8500 Å. In this way, the first encapsulation layer **201** not only may block the water vapor and oxygen for the display substrate **101**, but also does not affect the light extraction effect of the display panel **1**. Moreover, the manufacturing cost is saved. In addition, since industrial production efficiency of manufacturing the silicon oxynitride layer by using the CVD equipment is high, a purpose of improving the production efficiency may be achieved by using the CVD equipment to manufacture the silicon oxynitride layer.

The second encapsulation layer **202** may be made of a silicon nitride (SiNx) material. Since the second encapsulation layer **202** made of the silicon nitride material is intact and dense, the second encapsulation layer **202** may block the water vapor and oxygen for the display substrate **101**, and prevent the water vapor and oxygen from entering the electroluminescent devices and the base substrate **300**.

The embodiments of the present disclosure do not limit a specific value of a thickness of the second encapsulation layer **202**. As shown in FIG. 7, the thickness H2 of the second encapsulation layer **202** may be in a range of 1000 Å to 10000 Å, inclusive (i.e., $1000 \text{ Å} \leq H2 \leq 10000 \text{ Å}$). For example, the thickness H2 of the second encapsulation layer **202** may be 3000 Å, 5000 Å, 8000 Å, or 9000 Å. In this way, the second encapsulation layer **202** not only may block the water vapor and oxygen for the display substrate **101**, but also does not affect the light extraction effect of the display panel **1**. Moreover, the manufacturing cost is saved. In addition, since industrial production efficiency of manufacturing the silicon nitride layer by using the CVD equipment is high, the purpose of improving the production efficiency may be achieved by using the CVD equipment to manufacture the silicon nitride layer.

The third encapsulation layer **203** may be made of a silicon carbonitride material. The silicon carbonitride (SiCN) layer mainly plays a role of covering particles on the second encapsulation layer **202** and planarizing the second encapsulation layer **202**. The particles mainly come from an evaporation process and a transport process in manufacturing. The silicon carbonitride has the inorganic property, and thus has a good blocking property.

In some embodiments, the third encapsulation layer **203** may be made of a silicon carbide (SiC) material. Since the silicon carbide (SiC) has excellent flatness and is well bonded to the color filter layer **121**, the silicon carbide layer may be used as both a planarization layer and an adhesive layer. Therefore, in the case where the third encapsulation layer **203** is made of the silicon carbide (SiC) material, compared with the structure in FIG. 6, the adhesive layer **122** may be omitted.

Moreover, since the industrial production efficiency of the silicon carbonitride layer or the silicon carbide layer manu-

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factured by the CVD equipment is high, the purpose of improving the production efficiency may be achieved by using the CVD equipment to produce the silicon carbonitride layer or the silicon carbide layer.

The embodiments of the present disclosure do not limit a specific value of a thickness of the third encapsulation layer **203**. As shown in FIG. 7, the thickness H3 of the third encapsulation layer **203** may be in a range of 2000 Å to 20000 Å, inclusive (i.e., $2000 \text{ Å} \leq H3 \leq 20000 \text{ Å}$). For example, the thickness H3 of the third encapsulation layer **203** may be 3000 Å, 5000 Å, 8000 Å, 10000 Å, 15000 Å, or 18000 Å. In this way, the third encapsulation layer **203** not only may block the water vapor and oxygen for the display substrate **101**, but also does not affect the light extraction effect of the display panel **1**. Moreover, the manufacturing cost is saved.

The fourth encapsulation layer **204** may be made of the silicon carbonitride (SiCN) material. Compared with a parylene material, the silicon carbonitride material has the inorganic property. Thus, the fourth encapsulation layer **204** meets requirements for being bonded to the glass cover plate **3** located thereon. In this case, compared with the structure in FIG. 6, the third inorganic layer **15** may be omitted.

As shown in FIG. 7, a thickness H4 of the fourth encapsulation layer **204** may be in a range of 1000 Å to 10000 Å, inclusive (i.e., $1000 \text{ Å} \leq H4 \leq 10000 \text{ Å}$).

In some embodiments, the fourth encapsulation layer **204** may also be made of the silicon carbide (SiC) material. The fourth encapsulation layer **204** plays a role of planarization, so as to cover particles generated in the manufacturing process, and prevent the entering of the water vapor and oxygen.

In the case where the third encapsulation layer **203** is made of the silicon carbonitride material, the fourth encapsulation layer **204** may be made of either the silicon carbonitride material or the silicon carbide material. Similarly, in the case where the third encapsulation layer **203** is made of the silicon carbide material, the fourth encapsulation layer **204** may be made of either the silicon carbonitride material or the silicon carbide material.

The fourth encapsulation layer **204** is also manufactured by the CVD equipment. Compared with the structure in FIG. 6, the fourth encapsulation layer **204** manufactured by the CVD equipment replaces the second organic layer **14** manufactured by the MLD equipment and the third inorganic layer **15** manufactured by the ALD equipment.

Thus, by manufacturing the encapsulation layer **200** on the display substrate **101**, it is possible to effectively prevent the water vapor and oxygen from entering the display substrate **101**, so as to avoid the aging of the light-emitting functional layer **420**, thereby prolonging the service life of the display apparatus **100**. For example, the encapsulation layer **200** includes the first encapsulation layer **201**, the second encapsulation layer **202**, the third encapsulation layer **203** and the fourth encapsulation layer **204** that are stacked on the display substrate **101**. Since the layers that constitute the encapsulation layer **200** are manufactured by deposition using the CVD equipment with high efficiency, it may reduce a waste of the equipment, and improve the mass production efficiency. In addition, a shadow of the layer deposited by the CVD equipment is smaller, which is easier to control. Moreover, the number of the layers of the encapsulation layer **200** is reduced, which may simplify the manufacturing process, reduce the production cost, and realize the thin and light design of the display apparatus **100**.

In some embodiments, in the case where the requirements for the service life and performance of the display apparatus

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100 are not high, the first encapsulation layer 201 may be omitted. By utilizing only the second encapsulation layer 202, the third encapsulation layer 203 and the fourth encapsulation layer 204, it is possible to effectively prevent the water vapor and oxygen from entering the display substrate 101. In this way, the process is simplified, and thus the cost is reduced. In some embodiments, in the case where the plurality of light-emitting devices 400 includes the light-emitting devices 400 for emitting red light, the light-emitting devices 400 for emitting green light, and the light-emitting devices 400 for emitting blue light, there is no need to provide a color filter layer 121 in the display panel 1, and thus the fourth encapsulation layer 204 may also be omitted.

In some embodiments, at least two of the first encapsulation layer 201, the color filter layer 121 and the fourth encapsulation layer 204 may be omitted. In this way, the process is further simplified, and thus the cost is further reduced. Moreover, the thickness of the display apparatus 100 is further reduced, thereby realizing the thin and light design of the display apparatus.

As shown in FIG. 8, some embodiments of the present disclosure provide a method for manufacturing a display panel. For convenience of description, descriptions will be made below by taking an example of the structure of the display panel 1 shown in FIG. 7. The method includes the following steps (S111 to S116).

In S111, a display substrate 101 is formed.

The step of forming the display substrate 101 includes: forming a base substrate 300, first electrodes 410, light-emitting functional layers 420 and second electrodes 430 in sequence. Thin film transistors 320 for driving light-emitting devices 400 are integrated in the base substrate 300. Film layers that constitute the thin film transistors 320 may be formed by photolithography processes. The thin film transistors in the base substrate 300 may be silicon-based thin film transistors.

In S112, a first encapsulation layer 201 is formed on the display substrate 101 through chemical vapor deposition.

The first encapsulation layer 201 is formed on the display substrate 101 through a chemical vapor deposition process. The first encapsulation layer 201 is made of a silicon oxynitride (SiON) material. As a first layer of the encapsulation layer 200, the silicon oxynitride (SiON) layer has good water resistance and low refractive index, which helps electroluminescent devices to emit light. A thickness H1 of the silicon oxynitride layer may be 3500 Å.

In S113, a second encapsulation layer 202 is formed on the first encapsulation layer 201 through chemical vapor deposition.

The second encapsulation layer 202 is formed on the first encapsulation layer 201 through a chemical vapor deposition process. The second encapsulation layer 202 is made of a silicon nitride material. As a second layer of the encapsulation layer 200, the silicon nitride (SiNx) layer may play a role of blocking water vapor and oxygen, so as to prevent the water vapor and oxygen from entering the display substrate 101. A thickness H2 of the silicon nitride layer may be 5000 Å.

In S114, a third encapsulation layer 203 is formed on the second encapsulation layer 202 through chemical vapor deposition.

The third encapsulation layer 203 is formed on the second encapsulation layer 202 through a chemical vapor deposition process. In some embodiments, the third encapsulation layer 203 may be made of a silicon carbide material. Since the silicon carbide (SiC) has excellent flatness and is well bonded to a color filter layer 121, the silicon carbide layer

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may be used as both a planarization layer and an adhesive layer. In addition, since a bonding force between the silicon nitride and the color filter layer 121 is smaller than a bonding force between parylene and the color filter layer 121, when the color filter layer 121 is developed, there will be no residue on the third encapsulation layer 203, which improves a display quality.

In S115, the color filter layer 121 is formed on the third encapsulation layer 203.

For the color filter layer 121 including filter units of three primary colors, the filter units of three primary colors may be formed through three photolithography processes.

In S116, a fourth encapsulation layer is formed on the color filter layer 121 through chemical vapor deposition.

The fourth encapsulation layer is formed on the color filter layer 121 through a chemical vapor deposition process. The fourth encapsulation layer 204 may be made of a silicon carbonitride material. The silicon carbonitride (SiCN) layer mainly plays a role of planarization. The silicon carbonitride has the inorganic property, and thus has a good blocking property. In addition, the silicon carbonitride layer may be well bonded to the glass cover plate 3 located thereon.

It will be noted that, in some embodiments, S112 may be omitted. That is, the encapsulation layer 200 may not include the first encapsulation layer 201.

In some embodiments, the third encapsulation layer 203 may also be made of the silicon carbonitride (SiCN) material. In this case, the third encapsulation layer 203 made of the silicon carbonitride (SiCN) material may cover particles, so as to play a role of planarization.

In some embodiments, in the case where the plurality of light-emitting devices 400 includes the light-emitting devices 400 for emitting red light, the light-emitting devices 400 for emitting green light, and the light-emitting devices 400 for emitting blue light, there is no need to provide the color filter layer 121 in the display panel 1, and thus S115 may be omitted.

The foregoing descriptions are merely specific implementations of the present disclosure, but the protection scope of the present disclosure is not limited thereto, any changes or replacements that a person skilled in the art could conceive of within the technical scope of the present disclosure shall be included in the protection scope of the present disclosure. Therefore, the protection scope of the present disclosure shall be subject to the protection scope of the claims.

What is claimed is:

1. A display panel, comprising:

a display substrate; and
an encapsulation layer, wherein

the display substrate includes:

a base substrate; and
a plurality of light-emitting devices disposed on the base substrate, the plurality of light-emitting devices being configured to emit light; and

the encapsulation layer is located on a side of the light-emitting devices away from the base substrate, and is configured to encapsulate the light-emitting devices; the encapsulation layer includes:

a second encapsulation layer located on the side of the light-emitting devices away from the base substrate, and made of a silicon nitride material; and

a third encapsulation layer located on a side of the second encapsulation layer away from the light-emitting devices, and made of a silicon carbide material, wherein

the display panel further comprises a color filter layer located on a side of the third encapsulation layer away

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from the base substrate, and a fourth encapsulation layer located on a side of the color filter layer away from the third encapsulation layer, the fourth encapsulation layer being made of the silicon carbide material, wherein

the second encapsulation layer is in direct contact with the third encapsulation layer, the third encapsulation layer is in direct contact with the color filter layer, and fourth encapsulation layer is in direct contact with the color filter layer.

2. The display panel according to claim 1, wherein the encapsulation layer further includes:

a first encapsulation layer located between the second encapsulation layer and the light-emitting devices, and made of a silicon oxynitride material.

3. The display panel according to claim 2, wherein the first encapsulation layer is a first encapsulation layer manufactured by a chemical vapor deposition process;

the second encapsulation layer is a second encapsulation layer manufactured by another chemical vapor deposition process; and

the third encapsulation layer is a third encapsulation layer manufactured by yet another chemical vapor deposition process.

4. The display panel according to claim 1, wherein the base substrate is a silicon-based backplane integrated with a plurality of pixel circuits;

each of the light-emitting devices includes a first electrode, a light-emitting functional layer and a second electrode; and

the plurality of pixel circuits are electrically connected to respective first electrodes of the light-emitting devices.

5. The display panel according to claim 1, wherein the encapsulation layer further includes a first encapsulation layer located between the second encapsulation layer and the light-emitting devices, and made of a silicon oxynitride material;

the base substrate is a silicon-based backplane integrated with a plurality of pixel circuits;

each of the light-emitting devices includes a first electrode, a light-emitting functional layer and a second electrode; and

the plurality of pixel circuits are electrically connected to respective first electrodes of the light-emitting devices.

6. The display panel according to claim 5, wherein a thickness of the first encapsulation layer is in a range of 1000 Å to 10000 Å, inclusive;

a thickness of the second encapsulation layer is in a range of 1000 Å to 10000 Å, inclusive;

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a thickness of the third encapsulation layer is in a range of 2000 Å to 20000 Å, inclusive; and

a thickness of the fourth encapsulation layer is in a range of 1000 Å to 10000 Å, inclusive.

7. A display apparatus, comprising the display panel according to claim 1, and a glass cover plate configured to protect the display panel.

8. A method for manufacturing a display panel, comprising:

forming a display substrate;

forming, on the display substrate, a second encapsulation layer by a chemical vapor deposition process; and

forming, on the second encapsulation layer, a third encapsulation layer by another chemical vapor deposition process, wherein

after forming, on the second encapsulation layer, the third encapsulation layer by the another chemical vapor deposition process the method further comprises: forming a color filter layer on the third encapsulation layer; and forming, on the color filter layer, a fourth encapsulation layer by yet another chemical vapor deposition process;

the third encapsulation layer is made of a silicon carbide material, and the fourth encapsulation layer is made of the silicon carbide material; and

the second encapsulation layer is in direct contact with the third encapsulation layer, the third encapsulation layer is in direct contact with the color filter layer, and fourth encapsulation layer is in direct contact with the color filter layer.

9. The method for manufacturing the display panel according to claim 8, wherein before forming, on the display substrate, the second encapsulation layer by the chemical vapor deposition process, the method further comprises: forming, on the display substrate, a first encapsulation layer by yet another chemical vapor deposition process.

10. The method for manufacturing the display panel according to claim 8, wherein forming the display substrate includes: forming a silicon-based backplane integrated with a plurality of pixel circuits.

11. The method for manufacturing the display panel according to claim 9,

wherein the first encapsulation layer is made of a silicon oxynitride material; and

the second encapsulation layer is made of a silicon nitride material.

* * * * *